

Dyia QA Round 5 — Developer Response & Fix Report

Issued June 11, 2026 · Responds to: "DYIA QA – Test Round 5, v2" (May 18, updated Jun 10, 2026)

1. Executive Summary

All BUG-022 findings in Round 5 trace back to **one systemic root cause**: QA/branch test environments ran with **test-mode Stripe keys while writing to the production database**. This poisoned user rows with test-mode customer/subscription IDs that the live key cannot resolve, producing the "No such customer ... exists in test mode" errors, the "You already have a subscription" dead-ends, and several of the inconsistent tier displays.

We shipped: **(a)** self-healing billing routes, **(b)** a live/test **mode guard** that prevents this class of corruption permanently, **(c)** a fix for a state-machine hole that granted indefinite Pro access to expired trials, **(d)** aligned tier displays, and **(e)** a full reset of all 18 QA accounts so the subscription lifecycle can be re-tested cleanly. One item remains a **product decision** (Section 6).

2. Root Causes

Root cause	Mechanism	Permanent fix shipped
RC-1: Test/live mode mixing	Test-mode checkouts (QA branch envs) stored <code>cus_...</code> / <code>sub_...</code> ids from Stripe test mode in the shared DB; live-mode prod cannot read them. Test-mode webhook events could also rewrite rows matched by <code>stripe_customer_id</code> .	New <code>stripe_livemode</code> column stamps which mode created stored ids; all billing routes ignore wrong-mode ids; webhook drops events whose <code>livemode</code> doesn't match the server key. (Migration 044)
RC-2: DB-only subscription checks	Checkout blocked on the DB <code>subscription_status</code> field alone; portal used the stored customer id blindly. Stale/poisoned rows dead-ended users with no path to pay.	Stripe is now the source of truth: routes verify against live Stripe and self-heal stale pointers (clear + recreate / email re-link).
RC-3: Fail-open trial state	An expired trial <i>with</i> a subscription id on file stayed "trialing" forever if the trial-end webhook never landed → indefinite free Pro (account +102).	<code>computeSubscriptionState</code> now fails closed 24h after trial end when no status transition arrived.

3. Finding-by-Finding Response

Account / Finding	Reported symptom (Jun 10)	Diagnosis	Status
+101, +106, +107	"No such customer: 'cus_...'; a similar object exists in test mode, but a live mode key was used"	RC-1: row carried a test-mode customer id. Portal call with live key is guaranteed to fail.	FIXED self-heal + mode guard; accounts reset
+103, +104	"You already have a subscription. Please manage your existing billing..." when trying to upgrade	RC-2: DB said active/past_due but no live-mode subscription existed; route blocked without consulting Stripe.	FIXED Stripe-verified check; accounts reset
+102	Trial ended Apr 24; no payment ever made; still has Pro features on Jun 10	RC-3: fail-open trial state when the trial-end webhook never arrived.	FIXED fails closed after 24h grace; unit-tested
+103, +105, +107	"Pro" in Account → Subscription but "Basic" in header dropdown	Settings showed the <i>registered plan</i> (PRO) while the header showed <i>effective access</i> (Basic). Internally consistent but reads as a contradiction.	FIXED Settings now shows an explicit INACTIVE chip beside the plan badge
+100, +105	"Automatically updated to Basic with no payment" after failed payment	Working as designed: failed payment → 7-day dunning grace → Stripe auto-cancels → account downgrades to the free Basic experience. Not a defect; see Section 6 for the open product question.	BY DESIGN
Main concern	"Why pay for Basic if it remains available after a failed payment?"	Valid product gap: the post-failure fallback experience and the paid Basic plan are currently the same feature set.	PRODUCT DECISION (Section 6)

BUG-011, dunning emails	Pass	—	CLOSED
----------------------------	------	---	--------

4. Changes Shipped (code)

- Mode guard** — `src/lib/stripe-mode.ts` (new), migration `044_stripe_livemode_guard.sql`: stores which Stripe mode created the ids on each user row; checkout/portal/verify-session ignore wrong-mode ids; the webhook drops events whose `event.livemode` mismatches the server key and logs them to the admin webhook log.
- Self-healing portal** — `api/stripe/portal`: verifies the customer exists in the current mode, re-links by email when possible, otherwise clears poisoned pointers and returns a clear, actionable message instead of a raw Stripe error.
- Stripe-verified checkout** — `api/stripe/checkout`: "already subscribed" is now decided by live Stripe state, not the DB field; stale subscription pointers are cleared automatically (`resource_missing` → status reset to inactive).
- Trial fail-closed** — `src/lib/subscription.ts`: expired trials lose access 24h after trial end when no Stripe transition arrived (webhook-lag buffer prevents blips for legitimate conversions).
- Display alignment** — `Settings.tsx`: INACTIVE chip whenever a registered plan isn't currently paid/trialing, so Account and header agree.
- Tests** — 21 new unit tests covering the full QA matrix (dunning in/after grace, expired trials with/without subscription, canceled-with-time, scheduled downgrade, admin, Basic-not-Pro, AI credits) plus the mode-guard logic. Suite: 53 passing.

5. QA Accounts Reset

All 18 `maliarchuk.ann*` accounts were reset on Jun 11 to a clean never-subscribed state (Stripe pointers cleared, plan/status cleared, trial eligibility restored, dunning stamps removed). Rows are intact — `jobs/quotes/data` untouched. Re-runnable via `scripts/reset-qa-billing.mjs --apply`.

Heads-up for Round 6: please run billing tests against an environment whose Stripe mode matches its database (see Ops actions below). Testing test-mode checkouts against the production DB will re-create the corruption this round found.

6. Open Product Decision — post-failure Basic access

Today, a user whose payment fails (after the 7-day grace) keeps the full free Basic experience indefinitely, which makes the *paid* Basic plan economically equivalent to not paying. Options:

- Hard lock** after dunning grace: read-only access (view data, export) until payment resumes.
- Free tier split:** define a genuinely limited free tier (e.g., caps on jobs/quotes per month); paid Basic lifts the caps.
- Status quo**, treated as a deliberate retention/win-back funnel.

Engineering is ready to implement any of the three; awaiting product direction.

7. Ops Actions Required

#	Action	Owner	Status
1	Run <code>claudedocs/qa-round5-response/APPLY_qa_round5_migration.sql</code> in the Supabase SQL editor (adds <code>stripe_livemode</code> ; idempotent). Code is backward-compatible either way; the guard fully engages once applied.	Founder	TO DO
2	Deploy the fixes to production (merge + redeploy).	Founder	TO DO
3	Give QA/branch environments their own database (separate Supabase project) OR ensure any environment pointing at the prod DB uses the live Stripe key. The new mode guard contains the damage either way, but isolation is the real fix.	Founder	RECOMMENDED
4	In the Stripe dashboard, confirm only ONE webhook endpoint per mode points at production (<code>/api/stripe/webhook</code>), with the matching <code>whsec_...</code> in Vercel Production env.	Founder	VERIFY

8. Round 6 Re-test Checklist

1. **Fresh subscribe (Basic + Pro, monthly + annual):** each reset account → Settings → Account → Upgrade. Expect Stripe checkout with 14-day trial; tier badge + header agree after return.
2. **Portal:** "Manage Billing & Invoices" opens for every subscribed account — the test-mode customer error must not occur.
3. **Re-upgrade after failure/cancel:** cancel → wait for lapse (or failed card) → upgrade again. The "already have a subscription" dead-end must not occur unless Stripe really has a live subscription.
4. **Dunning:** fail a payment (Stripe test clocks or a declining card) → expect: failure email, banner countdown, Pro retained 7 days, locked after, INACTIVE chip in Settings.
5. **Trial expiry edge:** simulate webhook loss if possible — access must end ~24h after trial end regardless.
6. **Maps (new feature):** Maps tab visible in sidebar (Work group); pins render; pin detail panel (Open in Jobs / Directions / Call); Today-Week-Custom filters (Week/Custom are Pro); address autocomplete on the Job form stores a pin.

Dyia engineering — generated June 11, 2026. Companion files: [claudedocs/qa-round5-response/](#) (this PDF + migration), [scripts/reset-qa-billing.mjs](#), [claudedocs/billing-help/](#) (user-facing upgrade/downgrade screenshots).